

The chemical and structural origin of efficient p-type doping in P3HT

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ABSTRACT

We investigate the chemical and structural properties of solution-processed thin films of P3HT blended with p-type dopant F4TCNQ. The maximum in-plane electrical conductivity of doped films is observed at a molar doping fraction of 0.17, in agreement with the binding mechanism of F4TCNQ:P3HT complexes. Through the use of X-ray diffraction, a previously unreported crystalline phase is observed for P3HT films doped above a critical threshold concentration. This crystalline phase involves the incorporation of F4TCNQ molecules into ordered polymer regions and ultimately improves charge dissociation, leading to higher carrier density in thin film. Finally, optical absorption and X-ray diffraction reveal that the chemical state of P3HT in solution has a dramatic impact on the electrical and structural properties of the blended films.

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1. Introduction

Over the last decade organic semiconductors (OSCs) have been extensively investigated due to their tremendous potential for flexible electronics applications [1–4]. Recent advances in materials processing and design have led to field-effect mobilities exceeding those of amorphous silicon [5] and mass commercialization of organic electronics is coming close to fruition. As shown by the historic development of inorganic semiconductor devices [6], a necessary step to achieve the required engineering of electronic device properties is the realization of stable and controllable doped transport layers for both p- and n-type materials. Indeed significant improvements in the performance of organic light-emitting diodes (OLEDs) and organic photovoltaics (OPVs) have been reported when dopants are employed [7–11]. Doped OSC thin films have been shown recently to exhibit conductivities up to 5–8 orders of magnitude higher than undoped films [12,13]. A thorough

understanding of doped in OSCs will thus go a long way towards advancing the field of organic electronics.

Unlike doping in inorganic semiconductors, where host atoms are substituted by electron deficient (p-type) or electron rich (n-type) substitutes, doping in OSCs is achieved by the addition of a second species with appropriate molecular orbital energies leading to efficient ground-state charge transfer between OSCs and the dopant. Although charge transfer (CT) states can be observed in a variety of donor–acceptor blends [14–17], doping only occurs efficiently when the energy levels are correctly matched such that charge transfer occurs readily in the dark [18–20]. For p-type doping, the lowest unoccupied molecular orbital (LUMO) of the dopant should be close to or deeper than the highest occupied molecular orbital (HOMO) of the polymer. Ideally, each dopant molecule would transfer a full charge to the host semiconductor, and the charge would escape from the site of formation to become a free carrier. The increased carrier density not only gives rise to improved conductivities but also bleaching of the neutral polymer absorption that makes the material optically transparent in the visible [21,22].

For several years many have reported the efficient doping of OSCs using solution processable small molecules. In

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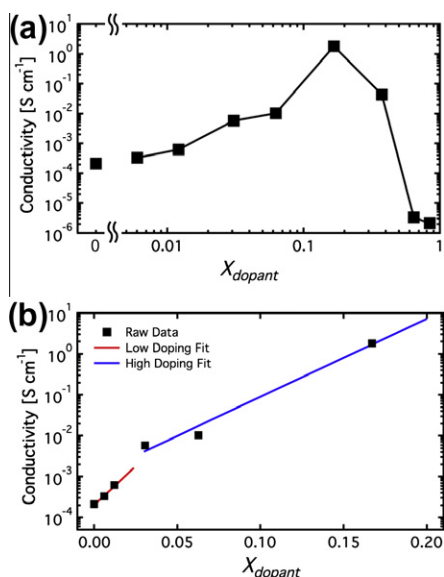


Fig. 1. (a) In-plane conductivities of P3HT films blended with F4TCNQ measured on ITO-patterned glass substrates as a function of dopant molar fraction. A maximum conductivity of 1.82 S cm^{-1} is obtained for F4TCNQ:P3HT blends at a dopant concentration of 0.17. (b) Fits to the conductivity data in the weak and strong doping regime.

this work we focus on the p-type doping of a prototypical, high performance semiconducting polymer poly(3-hexylthiophene) (P3HT) using 7,7,8,8-tetracyano-2,3,5,6-tetrafluoroquinodimethane (F4TCNQ). Due to a deep LUMO of 5.2 eV, F4TCNQ can efficiently undergo hole transfer to P3HT, which has a HOMO of $\sim 5.0 \text{ eV}$ [23,24]. Although several groups have successfully doped semiconducting polymers using F4TCNQ [25–27], there still lacks a general understanding of the doping process and how physical factors such as dopant dissolution in the casting solution and the final thin film, complex formation and crystalline structure affect doping. The manner in which dopant molecules are arranged in thin films is crucial to the efficacy of free-carrier formation and as such has a large impact on the overall conductivity of the doped system. In this report we elucidate how dopant complex formation in solution modulates the doping efficiency in P3HT films blended with F4TCNQ. We also reveal the formation of a previously unreported crystalline phase and comment on structural factors that limit the maximum achievable doping efficiency in OSCs. Our conclusions are general, however, and are indispensable in designing rational processing procedures for the fabrication of doped organic semiconductor layers. For example, the dopant choice, chemical state in solution and optimum doping concentrations can be tailored towards maximizing the conductivity enhancement in thin films.

2. Experimental

P3HT (Regioregularity > 97%, $M_w = 105 \text{ kDa}$, $M_n = 60 \text{ kDa}$) from Imperial College in London and F4TCNQ (Sigma–Aldrich) were used as received without further

purification. Films of pure P3HT and F4TCNQ as well as blended films were prepared on a variety of substrates depending on the application. All substrates were cleaned by sonication in acetone, methanol and isopropyl alcohol, rinsed with deionized water and blown dry. Samples were then exposed to a UV-Ozone environment for 20 min. Films were spin-cast from solutions at 80°C with different mass ratios of F4TCNQ to P3HT from chlorobenzene at 1000 rpm for 1 min and at 3000 rpm for 5 s. The concentration of all solutions is 2 mg solute per 1 mL of solvent. Thus a 50:50 F4TCNQ:P3HT solution (dopant molar fraction of 0.376) would contain 1 mg of F4TCNQ and 1 mg of P3HT per 1 mL of chlorobenzene.

Blended films of F4TCNQ and P3HT were spin-cast onto glass substrates with pre-patterned indium–tin–oxide contacts that define devices with a channel length of 5 mm and a channel width of 20 mm. Conductivity measurements were carried out in the dark and under vacuum ($\sim 10^{-3} \text{ mbar}$). Film thicknesses are determined using Atomic Force Microscopy. UV–Visible absorption spectra are collected for films spin-cast on plain glass substrates using a Varian Cary 6000i UV–Visible–NIR spectrophotometer. Films for GIXD experiments are fabricated by spin-casting onto heavily p-doped Si (100) substrates. Diffraction patterns are collected using an incident beam of 12.7 keV at a grazing angle of 0.1° and expressed as a function of the scattering vector $q = 4\pi \sin(\theta)/\lambda$. Here θ represents half of the scattering angle, λ is the wavelength of the incident beam, q_{xy} is the component of the scattering vector parallel to the substrate plane and q_z is the component perpendicular to the substrate plane. Note that the out-of-plane diffraction data presented in the manuscript do not correspond exactly to q_z since the pattern was collected in grazing incidence.

3. Results and discussion

We first analyze the effect of dopant concentration on electrical conductivity in spin-cast, blended films of P3HT with F4TCNQ on ITO-patterned glass substrates. Our results, summarized in Fig. 1a, show that adding F4TCNQ increases the in-plane conductivity of P3HT thin films by about four orders of magnitude up to a concentration of 17 dopant molecules per 83 monomer units (stated below as a dopant molar fraction or concentration of 0.17). The in-plane conductivity peaks at $1.82 \pm 0.22 \text{ S cm}^{-1}$ and decreases upon further addition of dopant. Analysis of the concentration-dependent conductivity, however, reveals three distinct transport regimes in F4TCNQ:P3HT blended films. At doping levels below 0.17, the conductivity exhibit two different regimes of exponential-like dependence on concentration: one at and below ~ 0.012 dopant concentration and the other from ~ 0.031 to ~ 0.17 . Above these concentrations, the conductivity quickly decreases as a function of F4TCNQ molar fraction. We will later show that these three regimes are correlated with thin film and solution absorption spectra as well as the crystalline structure in the solid state. Fig. 1b shows fits to the conductivity data demonstrating the difference in exponential dependence between low and high dopant concentrations, respectively.

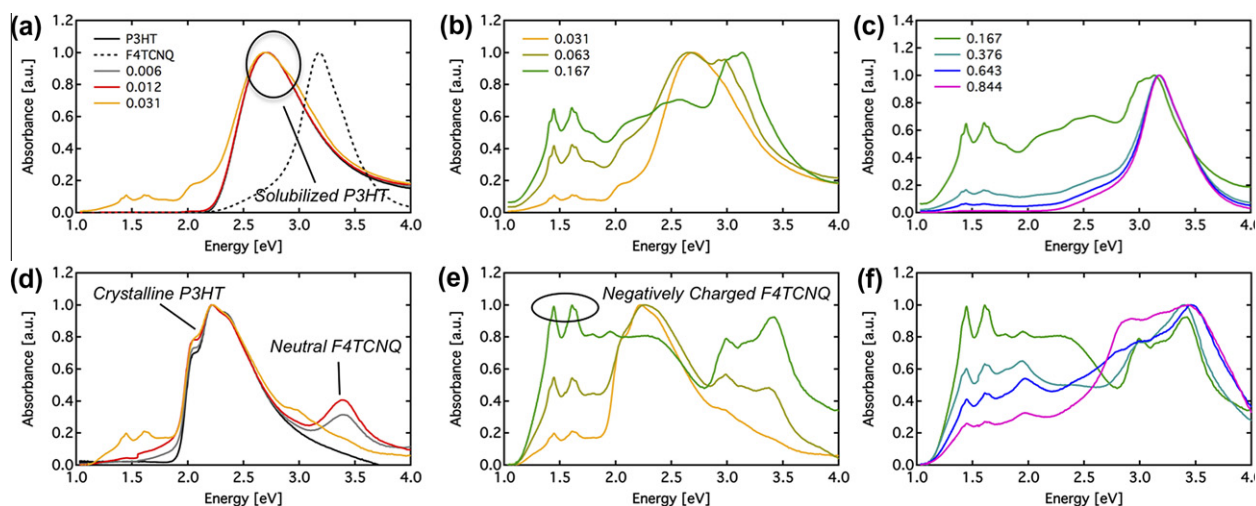


Fig. 2. Optical absorption spectra for F4TCNQ-doped P3HT solutions (a–c) and films (d–f) of different molar fractions revealing three distinct regimes of dopant concentrations.

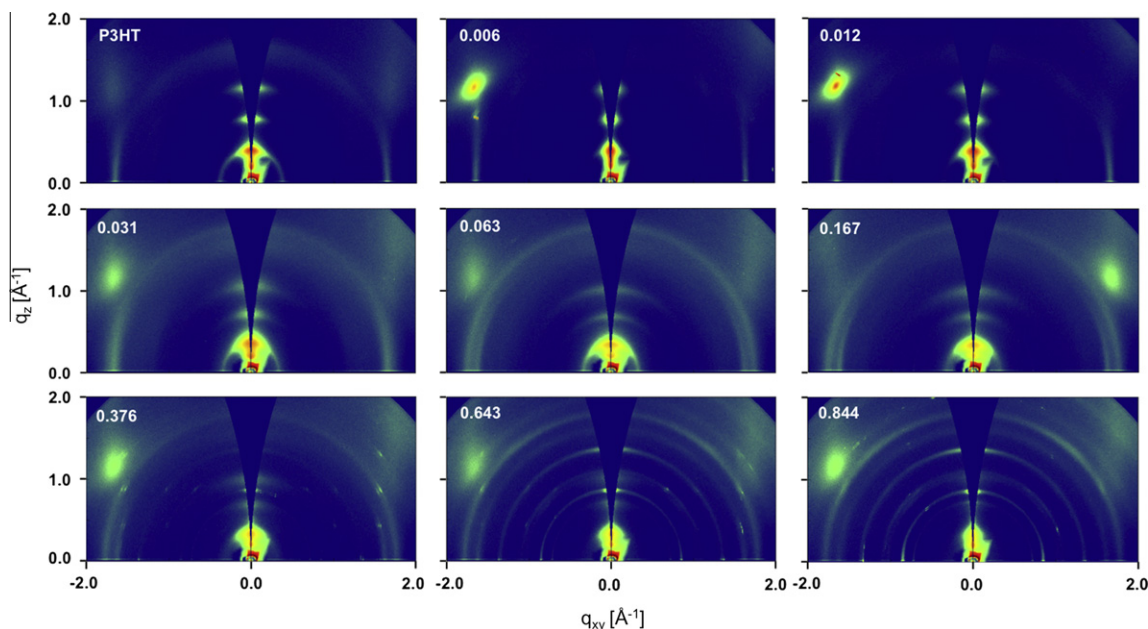


Fig. 3. 2D GIXD patterns for neat and blended films of P3HT with F4TCNQ at different doping concentrations.

In order to rationalize these trends in electrical performance, we have investigated structural properties of doped P3HT films using a combination of optical absorption spectroscopy and grazing incidence X-ray diffraction (GIXD). UV–Visible absorption spectra of F4TCNQ:P3HT solutions and spin-cast films are shown in Fig. 2, while the 2D diffraction patterns (Fig. 3) and the out-of-plane and in-plane cuts (Fig. 4) reveal the structural aspects of the doping process. The results can be broken down into three different regimes. For F4TCNQ:P3HT blends with dopant concentrations of ~ 0.012 or lower, the solution spectra mirror that of pure P3HT, which displays a single broad peak at ~ 2.7 eV. The resultant films exhibit vibronic

features between 2 and 2.5 eV and a broad peak centered at ~ 3.3 eV corresponding to ordered P3HT domains and neutral F4TCNQ molecules, respectively (Fig. 2a and d). Note that pure F4TCNQ does exist in solution and exhibits an absorption maximum at ~ 3.2 eV. However, at these low concentrations it is masked by the broad P3HT absorption and cannot be discerned in the spectrum (Fig. 2a). The crystalline structure of the films also closely resembles that of neat P3HT with indexed (100) and (010) peaks at $0.38 \text{ \AA}^{-1}$ and $1.64 \text{ \AA}^{-1}$ corresponding to a lamellar spacing of $16.5 \text{ \AA}$ and a π -stacking spacing of $3.83 \text{ \AA}$.

In the second doping regime, at a dopant molar fraction of 0.031, sub-bandgap peaks at ~ 1.43 eV and ~ 1.6 eV

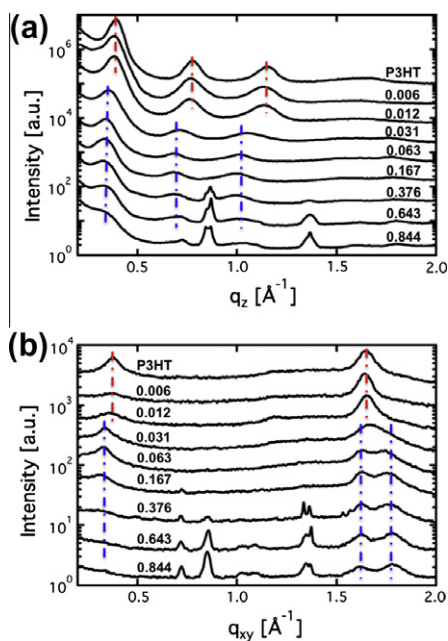


Fig. 4. Out-of-plane (a) and in-plane (b) diffraction patterns of F4TCNQ-doped P3HT thin films illustrating the formation of a new phase above a critical dopant concentration of ~ 0.03 . Red lines mark peaks belonging to crystalline P3HT and blue lines mark those belonging to the dopant-polymer crystalline phase. Narrow peaks that appear at high dopant concentrations belong to neutral F4TCNQ crystals. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

appear in the spectra of both film and solution and are attributed to dopant anions [28]. These sub-bandgap absorption features confirm the existence of a previously reported charged complex [25] and continue to increase in intensity with respect to pure P3HT absorption up to a dopant concentration of ~ 0.17 . The crystal structure of doped films above this critical threshold of 0.031 also changes entirely. The lamellar spacing increases to a value of $18.5 \text{ \AA}$ and there now exist two different in-plane d-spacings of $3.85 \text{ \AA}$ and $3.59 \text{ \AA}$. It is interesting to note that in the 0.031 dopant film, although the alkyl spacing changes from $16.5 \text{ \AA}$ to $18.5 \text{ \AA}$, the in-plane diffraction pattern exhibit a broad peak at $q \sim 1.65 \text{ \AA}^{-1}$ that seems to correspond to a mixture of neat P3HT and doped P3HT. Finally, as the concentration increases beyond 0.17, the absorption of ionized dopant decreases with respect to pure F4TCNQ absorption, which is characterized by features between 3 and 3.5 eV. In addition diffraction from pure F4TCNQ is observed in Figs. 3 and 4.

These results suggest that doping of P3HT via solution processing involves an intricate interplay between dopant dissolution and charge-transfer chemistry. When a dopant molecule is introduced into a polymer solution, it either remains solvated or attaches itself to the polymer backbone to form a charge-transfer species.



At low dopant concentrations, the solution absorption spectra show no sub-bandgap absorption, which implies that the dopant molecules are mostly free and have not

formed the charged complex (Fig. 2a). The resultant films also exhibit mostly pure absorption of the two materials and crystalline structures similar to that of neat P3HT. This suggests that most of the dopants are neutral and reside in the amorphous regions of the polymer film. However, the fact that there is an increase in conductivity infers that some dopants do form charged complexes in thin film. This is apparent by the presence of ionized F4TCNQ absorption that is several orders of magnitude lower than P3HT absorption and is detected using Photothermal Deflection Spectroscopy (PDS), an ultra-sensitive optical absorption spectroscopy, even at doping concentrations well below 0.01 (Fig. 5). For more details regarding PDS refer to Ref. [29]. We call this regime the “weak doping regime”.

When the F4TCNQ molar fraction is sufficiently high (>0.03), F4TCNQ:P3HT charged complexes are observed to form in solution. The two ionized species remain bound during thin film formation and eventually lead to the growth of a new semicrystalline phase. Ionized F4TCNQ (polaron) absorption is clearly observed beyond this threshold and the conductivity continues to increase as a function of dopant concentration. We call this regime the “strong doping regime”. It is important to note that the fraction of ionized dopant increases dramatically when the “strong doping regime” is reached, as is evident by the relative peak intensities between neutral and ionized F4TCNQ molecules in both thin film and solution. In order to accurately quantify this fraction, however, rigorous peak analyses and considerations of P3HT polaronic and bipolaronic absorption need to be accounted for. The ionization probability is currently under investigation.

We now compare the doping efficiency between the weak and strong doping regimes. Due to the nature of the exponential form of the concentration-dependent conductivity, however, it is not accurate to directly compare the slope of the two fits in a semi-log plot as in Fig. 1b (i.e. the derivative of an exponential form is dependent on more than the dopant concentration). In order to estimate the doping efficiency in the two regimes, we use a simple approximation:

$$\frac{d\sigma}{dc} \approx \frac{\Delta\sigma}{\Delta c} \quad (2)$$

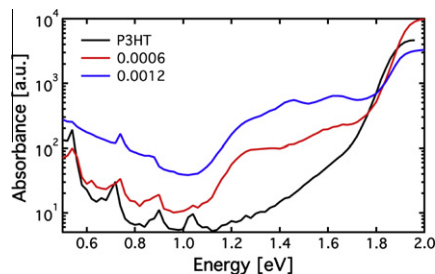


Fig. 5. Photothermal Deflection Spectroscopy (PDS) spectra for F4TCNQ:P3HT blended films drop-cast from 1,2-dichlorobenzene onto quartz substrates. As suggested by the increase in sub-bandgap absorptions at 1.43 eV and 1.6 eV, ionized F4TCNQ molecules are present even for dopant concentrations well below 0.01.

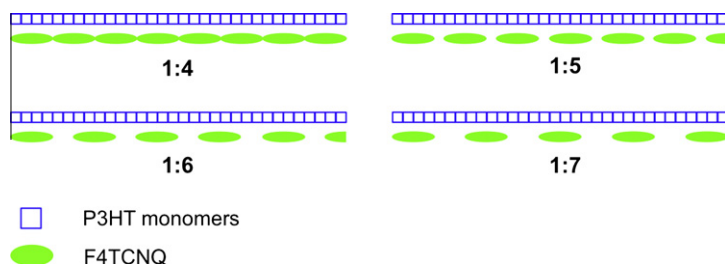


Fig. 6. Illustration of the various ways in which dopants can saturate polymer chains. Since each F4TCNQ molecule binds to approximately four P3HT monomers, in a saturated chain two adjacent F4TCNQ molecules can be separated by 0–3 monomers as demonstrated above. If all four cases occur with the same probability then there will be on average ~ 19 bounded dopants per 100 monomers (dopant concentration of ~ 0.16).

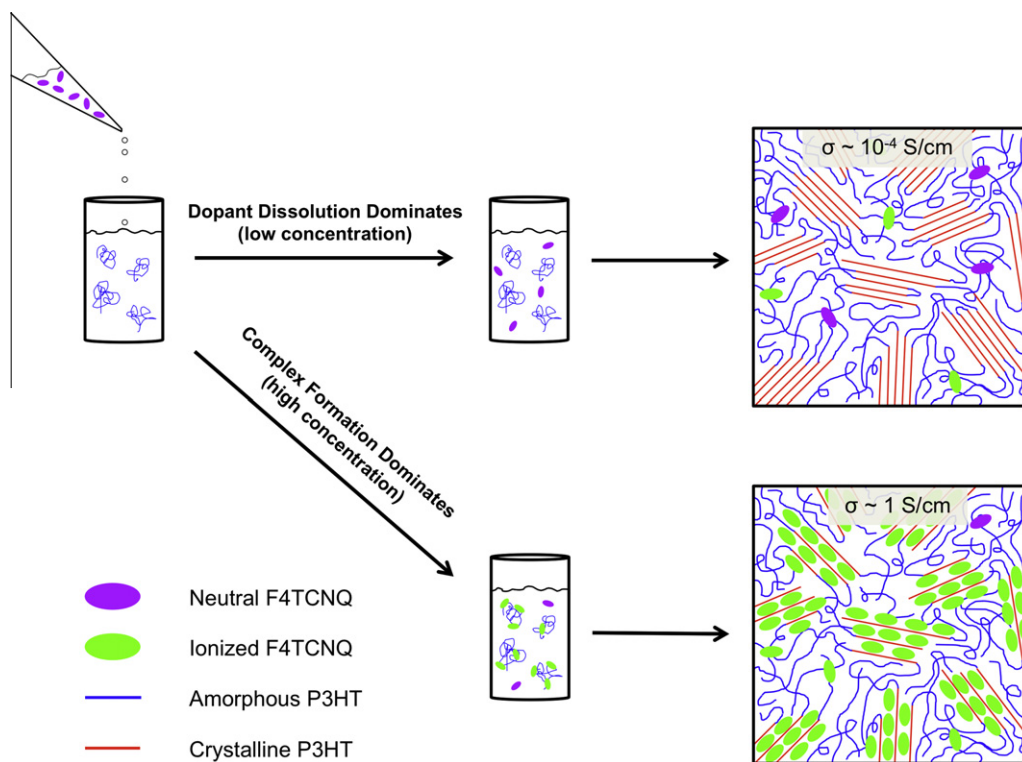


Fig. 7. Illustration of the dependence of doping efficiency in P3HT thin films on the solubility of F4TCNQ and its ability to form charged complexes in solution. At low dopant concentrations, F4TCNQ prefers to remain solvated and do not readily transfer a charge to P3HT. As a result F4TCNQ molecules reside in amorphous polymer regions in thin films and the conductivity is minimally improved in this “weak doping regime”. At high dopant concentrations, charged complexes form in solution and lead to the nucleation of dopant-incorporated crystalline polymer domains in thin film. The conductivity increases by up to four orders of magnitude in this “strong doping regime”.

Here we take the difference in conductivity between the two endpoints of each regime (0–0.012 for the weak doping regime and 0.031–0.17 for the high doping regime) and divide these by the difference in added dopants. We obtain doping rates of $3.32 \times 10^{-2} \text{ S cm}^{-1}$ and 13.3 S cm^{-1} per added mole fraction of dopant for the weak and strong doping regime, respectively. Thus the average doping efficiency is 2–3 orders of magnitude higher in the strong doping regime as compared to the weak doping regime.

As the concentration further increases, the amount of ionized dopants also increases and reaches a maximum at a doping concentration of 0.17. This upper limit in both

conductivity and low energy absorptivity ($\sim 1\text{--}1.7 \text{ eV}$) can be understood by investigating the nature of the dopant–polymer interaction. Pingel et al. have previously shown that a single F4TCNQ molecule interacts with approximately four thiophene monomers in order to form the charged complex [30]. If we assume that the complex formation process occurs in a random manner then, at saturation, two F4TCNQ molecules can be separated by anywhere between zero to three monomers. Therefore, the average *maximum* number of F4TCNQ molecules that can bind to a single polymer chain is approximately 19 in 100 monomers, or a doping concentration of ~ 0.16 (see Fig. 6).

Increasing the doping concentration to a value higher than this saturation point results in the formation of pure F4TCNQ crystals since there no longer exist free polymer chain segments where F4TCNQ can form complexes. Furthermore the volume fraction of conducting polymer chains decreases which leads to lower conductivity. From this analysis we can conclude that the size of the dopant molecule can significantly limit the maximum achievable doping density. This may be one of the reasons why P3HT films can exhibit conductivities of up to 30 S/cm when smaller dopants such as ClO_4^- and tridecafluoro-1,1,2,2-tetrahydrooctyl)trichlorosilane (FTS) are employed [22,31].

Although determining the crystal structure of the doped phase requires more peaks than we observe and is outside the scope of this report, we propose that the new crystalline phase involves the incorporation of F4TCNQ molecules into ordered polymer domains as evident from GIXD results. The measured π -stacking spacing of 3.59 Å is consistent with simulation results reported by Aziz et al. [32] where an optimal distance of 3.53 Å between the cyanide functional group on F4TCNQ and four thiophene monomers is observed. As previously stated, the average doping efficiency in the strong doping regime is 2–3 orders of magnitude higher than the weak doping regime. Calculations by Mityashin et al. [33] show that increasing the spatial density of doped molecules promotes higher electrostatic energetic disorder and increases the number of charge dissociation pathways. For our system, by forming the new crystalline phase the dopants molecules are no longer dispersed across (most likely) amorphous regions of the polymer films but rather aggregate to form densely packed dopant–polymer crystallites. Hence, the increase in doping efficiency that accompanies the formation of the new phase may be explained in terms of the creation of charge dissociation pathways in agreement with the reported calculations [33]. This mechanism is expected to greatly enhance electrical conductivity due to the fact that charges dissociate more easily and the carriers are generated directly into percolating crystalline domains where charge transport occurs. These data convincingly show that the ability of the dopant to form charged complexes in solution ultimately controls the doping efficiency in thin film, as summarized by Fig. 7.

4. Conclusion

The ability to efficiently dope organic semiconductors is essential for the advancement of the field of organic electronics and only by better understanding the doping process can new materials be designed and can rational processing procedures be devised. In this work we reveal that for a particular dopant there is a maximum achievable doping efficiency that is limited by both the size of the dopant and the manner in which it binds to the host semiconductor. Secondly, we investigated the interplay between dopant dissolution and complex formation and show that doping occurs efficiently when P3HT:F4TCNQ charged complexes form in solution. The dissolution of the dopant in solution hinders charged complex formation and thereby reduces film conductivity. Finally, we present

experimental evidence for improved charge carrier dissociation via the formation of densely packed dopant–polymer crystallites. Our results elucidate the formation of doped OSC thin films and, by revealing the role of solvation properties and molecular structures in modulating doping efficiency, we provide key insights for the future design and processing of organic semiconducting dopants.

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